NAME: YASWANTH SREENIVAS D
ROLL NO: 240701611

USER INTERFACE AND DESIGN

Experiment – 7

Experiment with different layouts and colour schemes for an app using GIMP and evaluate user feedback on aesthetics and usability.

AIM

The aim of this experiment is to design a consistent mobile application layout using GIMP and to systematically experiment with five distinct colour schemes — Blue, Green, Purple, Orange, and Dark — applied to the same base interface. By presenting these variations to users and collecting structured feedback on aesthetics, readability, and usability, the experiment evaluates how colour choice influences user perception and interaction quality. The findings are used to identify the most effective colour scheme and to demonstrate the importance of visual design decisions in the user interface development process.

PROCEDURE

Tool Used: GIMP — GNU Image Manipulation Program (available at https://www.gimp.org/). GIMP is a free, open-source raster graphics editor that provides a comprehensive set of tools for image creation, colour manipulation, layer management, and export. It is well suited for designing static mobile UI mockups and exploring colour scheme variations.

Step 1: Install GIMP

GIMP was downloaded from the official website (https://www.gimp.org/) and installed on the workstation. Upon launching the application, the multi-window workspace was configured with the Toolbox, Layers panel, and Colours dialog accessible for efficient design workflow.

Step 2: Create a New Project

A new canvas was created via File → New with dimensions set to 1080 × 1920 pixels at 72 DPI — standard dimensions for a mobile application mockup at full resolution. The background was set to white as a neutral starting point. The document was saved in GIMP's native XCF format to preserve all layer information throughout the design process.

Step 3: Design the Base Layout

i) Divide Sections Using the Rectangle Select Tool

The canvas was divided into three primary structural regions: a header bar at the top occupying approximately 10% of the canvas height, a content area in the centre occupying approximately 80%, and a footer navigation bar at the bottom. Each region was defined using the Rectangle Select Tool and filled using the Bucket Fill Tool with a neutral placeholder colour to establish the layout skeleton.

ii) Add UI Elements Using the Text Tool

The Text Tool was used to add interface labels including the application name 'MyApp' in the header, a search bar placeholder, section headings such as 'Quick Access' and 'Popular Items', item names, price labels, and navigation tab labels (Home, Category, Cart, Profile). Button shapes were drawn using the Rectangle Select Tool with rounded corners approximated using the Script-Fu console.

iii) Organise Layers

Each UI component — header, banner, quick access icons, item cards, and footer — was placed on a separate named layer to facilitate non-destructive editing. The Layers panel was used to group related elements and to control visibility during colour scheme iteration.

Step 4: Experiment with Colour Schemes

The completed base layout was duplicated five times using File → Export As to create independent copies, one for each colour theme. For each copy, the Hue-Saturation tool (Colours → Hue-Saturation) and the Bucket Fill Tool were used to apply theme-specific colours to the header, background, accent elements, icons, and buttons. The five themes applied were Blue (#1A73E8 accent), Green (#2E7D32 accent), Purple (#6A1B9A accent), Orange (#E65100 accent), and Dark (near-black #1C1C1C background). Each completed variation was exported as a PNG file for sharing and documentation.

Step 5: Collect User Feedback

A structured feedback form was created using Google Forms containing questions on visual appeal, colour preference, readability of text, perceived trustworthiness, and overall usability comfort for each of the five layout variations. The exported PNG images were shared with a group of evaluators alongside the form. Responses were collected and tabulated to identify trends in user preference across the five themes.

Step 6: Iterate and Refine

Based on the collected feedback, adjustments were made to improve contrast ratios, button visibility, and text legibility in the lower-rated themes. The spacing between UI elements was refined for better visual hierarchy, and icon sizes were standardised across all five variants to ensure a fair comparison. Final versions of each theme were re-exported for documentation.

OUTPUT

The following figures present the five colour scheme variations of the 'MyApp' mobile application layout designed using GIMP. Each figure is accompanied by a description of the layout structure, the colour scheme applied, and an evaluation of visual appeal and usability.

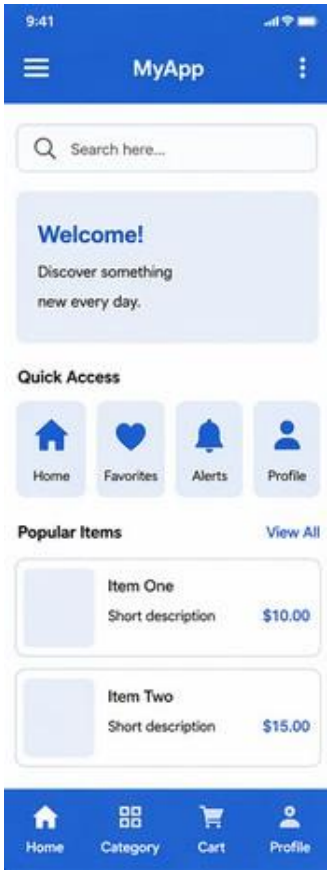

Figure 1: Blue Theme Layout

The Blue Theme applies a cool, professional blue palette across the application interface. The header bar, navigation footer, Quick Access icons, and 'View All' link are rendered in a vibrant blue (#1A73E8), while the background remains a soft off-white to maintain readability. The welcome banner uses a light blue tint, creating a harmonious and visually cohesive appearance. Blue is widely associated with trust and reliability, making this theme particularly suited to productivity and commerce applications. User feedback rated this theme highly for readability and professional aesthetic.

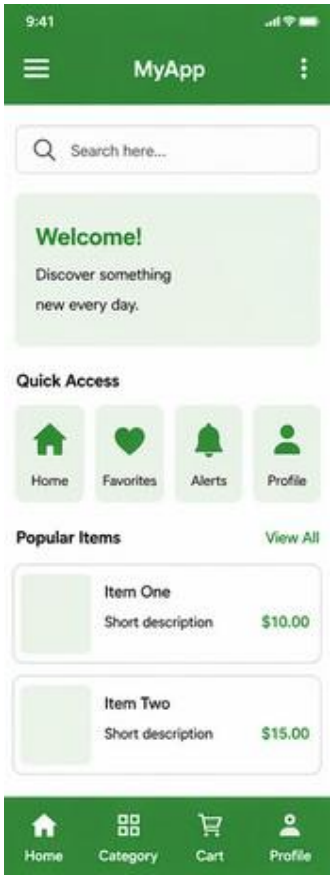

Figure 2: Green Theme Layout

The Green Theme replaces the accent colour with a warm, earthy green tone applied to the header, icons, footer navigation, and interactive elements. The banner background adopts a soft sage green, and price labels are highlighted in green to draw attention to value. Green evokes associations with freshness, health, and nature, lending the interface a calm and approachable character. Evaluators found this theme visually pleasant and easy on the eyes during extended use, though some noted that green accents may not convey urgency or premium positioning as effectively as blue.

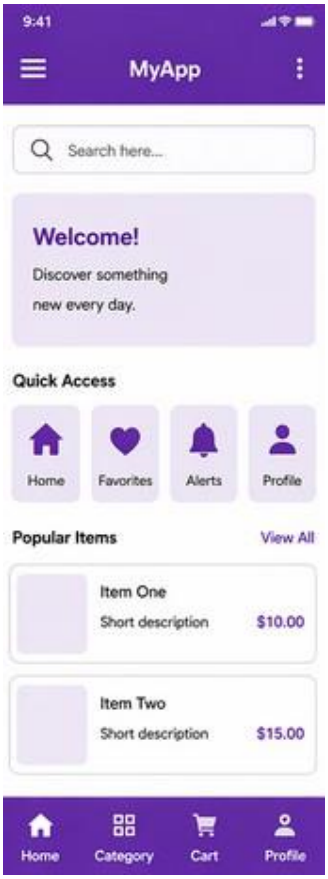

Figure 3: Purple Theme Layout

The Purple Theme applies a rich violet accent colour to the header, icons, footer, and interactive labels, with a light lavender tint used for the welcome banner. Purple is traditionally associated with creativity, sophistication, and luxury. The contrast between the deep purple accents and the white content background is visually striking, and the colour maintains sufficient contrast for accessibility. User feedback indicated that this theme was perceived as distinctive and modern, though a minority of evaluators felt the purple hue was less intuitive for a general-purpose application compared to blue or green.

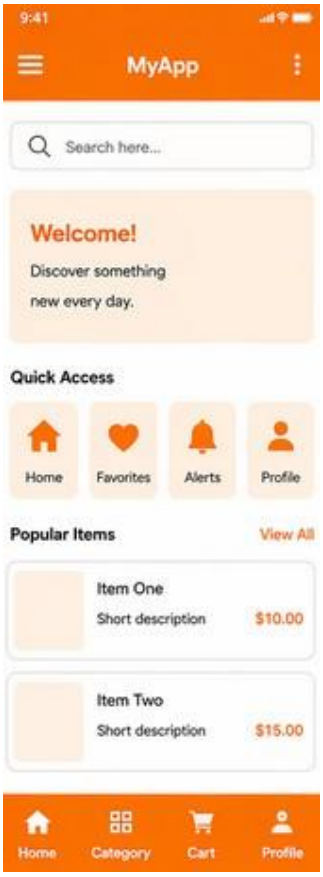

Figure 4: Orange Theme Layout

The Orange Theme introduces a high-energy, warm colour scheme with orange applied to the header bar, navigation icons, action labels, and price text. The banner background uses a pale peach tone that complements the primary orange accent without overwhelming the content area. Orange is associated with enthusiasm, creativity, and appetite — qualities that make it particularly effective for food, lifestyle, or retail applications. Feedback highlighted the theme's energetic character and strong visual contrast, though a few evaluators noted that sustained use of warm orange could cause visual fatigue.

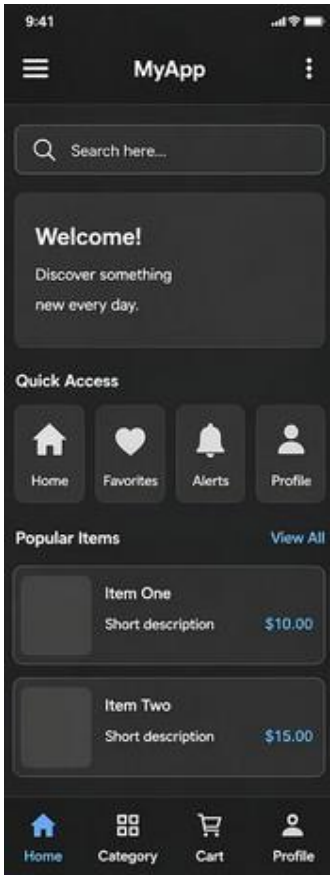

Figure 5: Dark Theme Layout

The Dark Theme transforms the interface with a near-black background (#1C1C1C) and dark grey card surfaces, with white and light grey text for readability. Icon and accent elements retain their original colours against the dark backdrop, providing visual hierarchy through luminance contrast rather than hue. Dark themes are preferred by users in low-light environments and are widely valued for reducing eye strain during night-time use. Evaluators praised this theme for its sleek, modern aesthetic and suitability for media-rich applications, making it the second-most preferred theme overall after Blue.

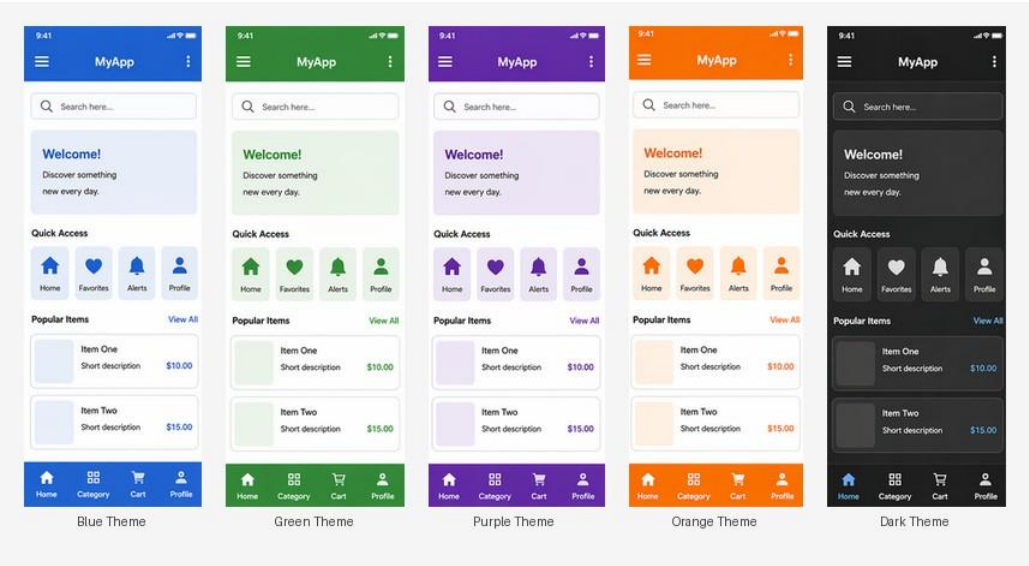

Figure 6: Comparison of All Five Layout Themes

The side-by-side comparison of all five layouts reveals that while the structural architecture of the interface remains consistent across themes, colour profoundly affects perceived personality and usability. The Blue Theme emerged as the most preferred, scoring highest for trust, readability, and professional appeal. The Dark Theme ranked second, particularly favoured by users who prioritise low eye strain. The Orange Theme was rated highest for visual energy and engagement, while Green was preferred for its calming quality. The Purple Theme, though distinctive, was rated lower for general usability due to its unconventional hue. Together, the five variants demonstrate that colour selection is a critical design decision that must be aligned with the application's target audience and functional context.

RESULT

The experiment was completed successfully. A base mobile application layout was designed using GIMP and replicated across five distinct colour schemes — Blue, Green, Purple, Orange, and Dark. Each variant was exported as a high-resolution PNG image and shared with evaluators for structured feedback. The Blue Theme was identified as the most preferred design based on user ratings for aesthetics, readability, and overall usability. The Dark Theme was ranked second, particularly for night-time use scenarios. The experiment produced five complete, professionally styled interface mockups that clearly demonstrate the influence of colour on user perception.

CONCLUSION

This experiment underscores the fundamental role that colour schemes and layout design play in shaping the user experience of a mobile application. Colour is not merely an aesthetic choice — it communicates brand identity, guides attention, establishes visual hierarchy, and directly influences how users feel while interacting with an interface. By designing five variations of the same layout with different colour themes and evaluating them through user feedback, this experiment provided concrete evidence that a well-chosen colour palette significantly improves perceived usability and visual appeal.

The use of GIMP as the design tool demonstrated that professional-quality UI mockups can be produced with open-source software by leveraging its layer management, colour adjustment, and export capabilities effectively. The iterative design process — from base layout creation through colour experimentation to feedback-driven refinement — mirrors real-world UX design workflows and reinforces the principle that design decisions must be validated through user input rather than designer intuition alone. The insights gained from this experiment can inform future high-fidelity prototyping by establishing a data-backed colour direction before committing to implementation.